

TrustSupport: An Evidence-Grounded AI Customer Support Agent

Abstract

TrustSupport is an AI customer-support prototype developed using the Customer Support on Twitter (TWCS) dataset, focusing on Amazon customer-support conversations. It classifies customer messages, retrieves historically relevant AmazonHelp responses, provides evidence-grounded responses, and decides whether to auto-handle or escalate cases to a human.

AmazonHelp conversations were extracted using customer-to-brand response relationships. Eight intents were defined: **Order & Delivery, Returns & Refunds, Payment & Amazon Pay, Account & Security, Product Problems, Prime Membership & Benefits, App / Device / Technical, and Prime Video / Streaming**. A manually labelled gold set of 200 examples was created for evaluation.

Two TF-IDF + Logistic Regression baselines were compared. The unweighted model achieved **30.0% accuracy and 0.06 macro-F1**, while the balanced model achieved **50.0% accuracy and 0.42 macro-F1**. Final held-out evaluation achieved **47.5% intent accuracy and 0.419 macro-F1**. The escalation classifier achieved **60.0% accuracy, 57.1% precision, 23.5% recall, and 0.333 F1**. The low recall highlights the importance of analysing escalation false negatives.

Response quality was evaluated using an **LLM-as-judge** rubric covering relevance, grounding, helpfulness, tone, hallucination, and overall quality. On 20 examples, the LLM judge achieved **80% exact agreement with human overall ratings**, with a **quadratic Cohen's kappa of 0.916**.

1. Introduction

Customer-support systems must do more than generate fluent text. A useful support agent should understand the customer's issue, provide a response consistent with the company's previous support behaviour, and recognize situations where automated handling may be inappropriate.

This project develops **TrustSupport**, an evidence-grounded customer-support prototype for Amazon. The system uses historical customer-support interactions from the TWCS dataset rather than relying only on free-form generation. This allows the system to use previous AmazonHelp responses as evidence when responding to similar customer questions.

The project defines “good” customer support as a combination of **relevance, grounding, helpfulness, appropriate tone, and safe escalation**. The prototype does not attempt to perform real-time order lookup, modify customer accounts, issue refunds, or perform transactional actions. Instead, it focuses on classification, historical retrieval, response grounding, and escalation.

2. Dataset and Problem Definition

The **Customer Support on Twitter (TWCS)** dataset contains approximately **2.8 million tweets** from customer-support interactions across multiple brands, including customer messages, brand responses, and tweet relationships.

AmazonHelp was selected as the target brand. Customer–brand interactions were reconstructed using `response_tweet_id`. This produced approximately **324,816 Amazon-related rows**, which were used to create customer–AmazonHelp response pairs for historical response retrieval.

The system solves three main tasks:

1. **Intent classification** — determine the customer's support intent.

2. **Historical response retrieval** — retrieve a relevant AmazonHelp response from previous interactions.
3. **Escalation decision** — determine whether the case should be automatically handled or sent to human support, with a reason.

3. Intent Definition and Gold Dataset

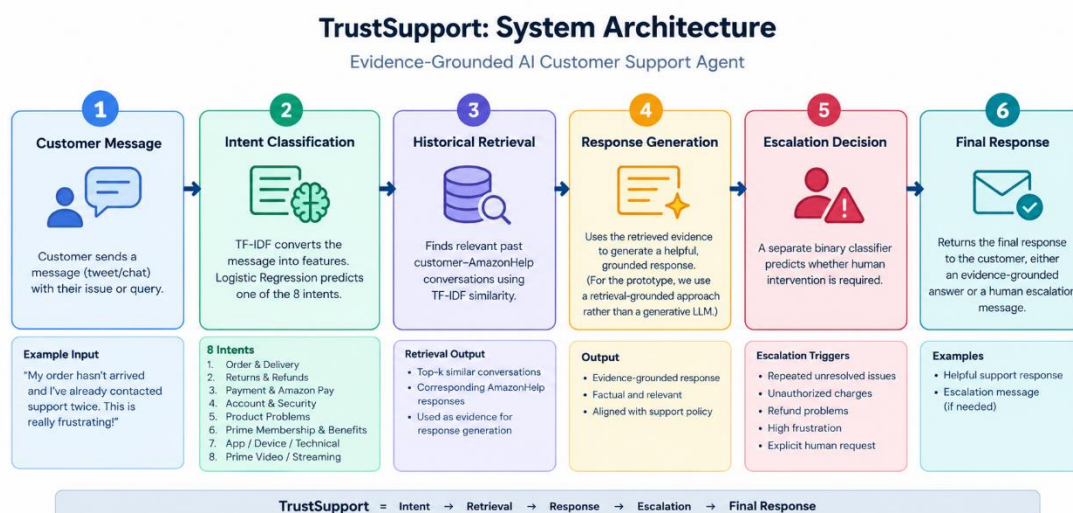
Eight intents were defined based on patterns discovered in Amazon customer messages:

Intent	Description
Order & Delivery	Delivery delays, tracking and missing orders
Returns & Refunds	Returns, cancellations and refund issues
Payment & Amazon Pay	Charges, payments and Amazon Pay problems
Account & Security	Account access and security concerns
Product Problems	Damaged, defective, fake or incorrect products
Prime Membership & Benefits	Prime membership and Prime-related benefits
App / Device / Technical	Application, Kindle and device-related problems
Prime Video / Streaming	Streaming, video and content issues

Repeated or unresolved customer-service complaints were treated as **escalation signals** rather than separate intents. A **200-example manually labelled gold dataset** was created, with intent, escalation, and escalation reasons. An **80/20 stratified split** produced **160 training and 40 held-out test examples**. Due to the small dataset size, results are considered **prototype-level measurements**.

4. System Architecture

Figure 1: System Architecture of TrustSupport



4.1 Intent Classification

TF-IDF converts customer messages into features, and Logistic Regression classifies them into eight intents. Two versions were evaluated: **unweighted** and **balanced** Logistic Regression. Class weighting was used to handle imbalanced intent classes.

4.2 Historical Response Retrieval

TF-IDF similarity retrieves relevant historical AmazonHelp responses. Exact self-matches were excluded during evaluation to avoid data leakage. On 20 examples, the average top-1 similarity was **0.523**, which is only a descriptive lexical-similarity measure.

4.3 Response Generation

For non-escalated cases, TrustSupport returns a relevant historical AmazonHelp response as evidence-grounded output. Escalated cases receive a human-support recommendation instead of an automated resolution.

4.4 Escalation

The escalation component identifies unresolved issues, repeated support failures, sensitive concerns, and strong complaints. It is evaluated separately because even a correctly classified intent may require human intervention.

5. Experimental Results

5.1 Intent Classification

Model	Accuracy Macro-F1	
Majority-class baseline	30.0%	≈ 0.058
TF-IDF + Logistic Regression	30.0%	0.060
Balanced TF-IDF + Logistic Regression	50.0%	0.420
Final held-out evaluation	47.5%	0.419

The unweighted model performed almost the same as the majority baseline. Class balancing improved macro-F1 from 0.06 to 0.42, showing that class imbalance significantly affected the initial model.

Figure 2: Confusion Matrix of Intent Classification on the Held-Out Test Set



The confusion matrix shows that some categories were recognized more reliably than others. Confusion occurred particularly between related support categories such as product problems, returns/refunds, payment issues, and account-related problems.

5.2 Escalation Results

On the 40-example held-out test set, the escalation classifier achieved:

- **Accuracy:** 60.0%
- **Precision:** 57.1%
- **Recall:** 23.5%
- **F1:** 0.333
- **False Negatives:** 13

The difference between accuracy and recall is important. Although 60% accuracy may appear reasonable, the classifier detected only 23.5% of the examples labelled as requiring escalation.

5.3 Response Quality and LLM-as-Judge

A structured LLM-as-judge rubric was created with six dimensions: **Relevance, Grounding, Helpfulness, Tone, Hallucination, and Overall quality**. Each dimension was rated from 1–5, and 20 examples were independently reviewed by a human using the same scoring framework to validate the judge. The LLM and human judges achieved:

Dimension	Exact Agreement	Quadratic Cohen's κ
Relevance	45%	0.545
Grounding	25%	0.418
Helpfulness	5%	-0.044
Tone	0%	0.015
Hallucination	10%	0.000
Overall	80%	0.916

The strong agreement on the overall score indicates useful consistency at the overall evaluation level. However, lower agreement for individual dimensions shows that subjective properties such as tone and helpfulness can be interpreted differently by human and LLM judges. Therefore, the LLM judge is treated as a structured supplementary evaluator rather than a replacement for human evaluation.

6. Failure Analysis

The evaluation identified five major failure modes.

6.1 Repeated or Unresolved Support Problems

The classifier sometimes failed to recognize that a customer had already attempted to obtain help. Statements indicating repeated failure or inability to contact support were sometimes predicted as non-escalation.

Hypothesis: The small TF-IDF training set does not contain enough linguistic variations of unresolved-support complaints.

6.2 Multi-Issue Messages

Some customer messages contain several problems at once, such as a damaged product combined with a failed refund request and unsuccessful customer-service contact.

Hypothesis: A single-label classifier struggles when multiple intents and escalation signals appear in the same message.

6.3 Indirect Frustration

Customers may communicate escalation-worthy situations using indirect expressions of dissatisfaction rather than explicit escalation words.

Hypothesis: TF-IDF relies heavily on surface-level word patterns and may not generalize well to different expressions of frustration.

6.4 Intent Errors Affecting Escalation

Several escalation false negatives were also associated with incorrect intent predictions. However, some examples had the correct intent while still failing to escalate, showing that escalation is a separate modelling problem.

6.5 Limited Gold Dataset

Only 200 manually labelled examples were available, producing a 40-example held-out test set. This limits the diversity of language available for learning escalation patterns.

7. Misleading Headline Metric

A misleading headline metric is **escalation accuracy**.

The escalation classifier achieved 60.0% accuracy, but its recall was only 23.5%, with 13 false negatives. These false negatives represent cases that were labelled as requiring human intervention but were predicted as safe for non-escalated handling.

Therefore, accuracy alone does not adequately describe the safety of the escalation component. **Escalation recall and false-negative analysis provide a more informative view of this component.**

Similarly, intent accuracy should be interpreted together with macro-F1 because the dataset is imbalanced. The majority-class baseline demonstrates that a model can obtain approximately 30% accuracy without effectively distinguishing all eight intents.

8. Decision Log

1. Selected AmazonHelp for its large support-interaction volume.
2. Used response_tweet_id to reconstruct customer–brand relationships.
3. Defined eight intents to keep classification focused.
4. Treated unresolved support as an escalation signal rather than an intent.
5. Created 200 hand-labelled examples for evaluation.
6. Used stratified splitting to handle class imbalance.

7. Compared unweighted and balanced Logistic Regression.
8. Selected the balanced classifier based on improved macro-F1.
9. Used TF-IDF retrieval for historical-response matching.
10. Excluded self-matches to prevent retrieval leakage.
11. Grounded responses in historical AmazonHelp replies.
12. Analysed escalation false negatives as important safety failures.
13. Validated the LLM judge against human ratings.
14. Used held-out evaluation for final results.
15. Reported limitations due to the small labelled dataset.

9. Limitations and Future Work

The main limitations are the small gold dataset of 200 examples and the 40-example held-out test set, which may cause metric variation. TF-IDF has limited semantic understanding, affecting retrieval and escalation for differently worded or multi-issue messages. The LLM-human response-quality evaluation was also limited to 20 examples, with lower agreement on some individual dimensions.

With one additional week, the system could be improved by:

- Expanding and diversifying the gold dataset.
- Using transformer-based intent classification.
- Introducing semantic/vector retrieval.
- Developing a higher-recall escalation mechanism.
- Adding explicit multi-intent detection.
- Increasing human and LLM response-quality evaluation.
- Testing retrieval and response quality on a larger held-out sample.

10. Conclusion

TrustSupport demonstrates an evidence-grounded approach to automated customer support using real historical customer-service conversations. The system combines intent classification, historical response retrieval, response grounding, and human escalation.

The experiments show that class imbalance substantially affects intent classification, with balanced Logistic Regression improving macro-F1 from 0.06 to approximately 0.42. However, escalation remains challenging: the classifier achieved 60.0% accuracy but only 23.5% recall, demonstrating why false-negative analysis is important.

The project also demonstrates the value and limitations of LLM-based evaluation. Overall LLM-human agreement was 80%, with a quadratic Cohen's kappa of 0.916, while individual dimensions showed considerably lower agreement.

Overall, the prototype provides a reproducible foundation for evidence-grounded customer support while clearly identifying the areas requiring further work before deployment in a real customer-service environment.